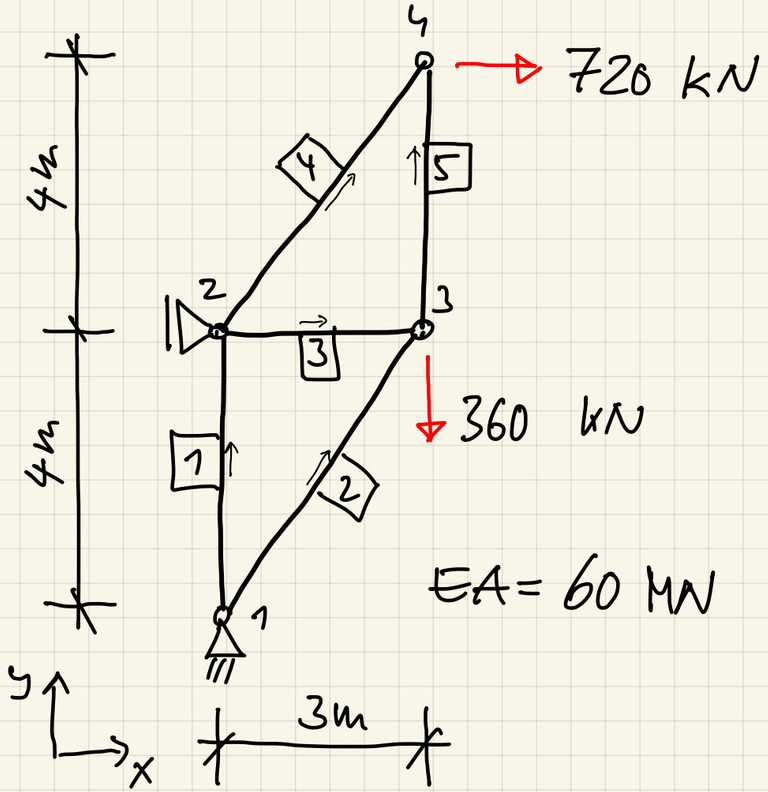
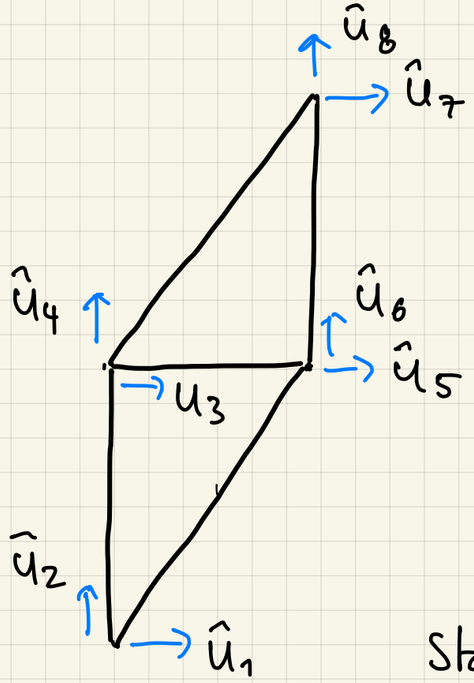
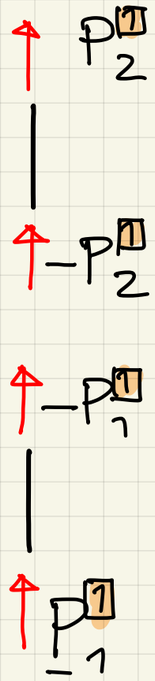
Finite Elemente für Stabtragwerke

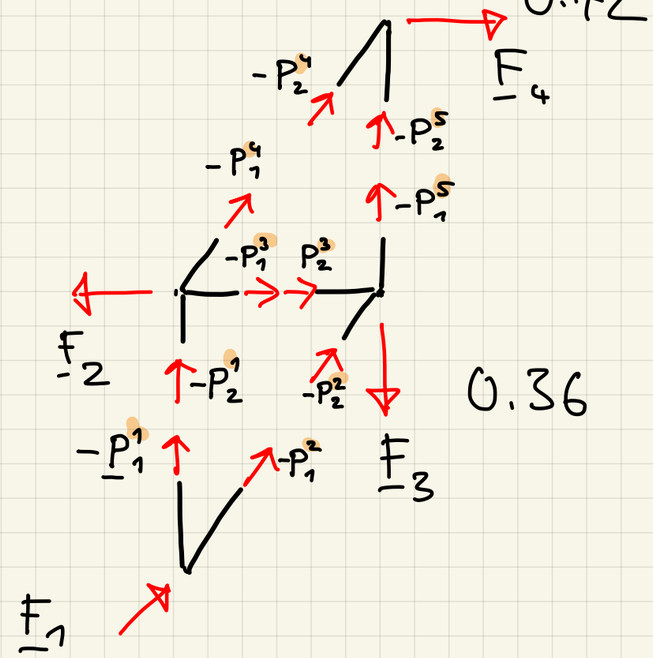
Fachwerke: Kräftegleichgewicht, Systemgrößen und Verschiebungen





Stab 1





$$\underline{\hat{u}} = \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$

$$\begin{aligned} \underline{P}_1^1 + \underline{P}_1^2 &= \underline{F}_1 \\ \underline{P}_2^1 + \underline{P}_1^3 + \underline{P}_1^4 &= \underline{F}_2 \\ \underline{P}_2^2 + P_2^3 + \underline{P}_1^5 &= \underline{F}_3 \\ \underline{P}_2^4 + \underline{P}_2^5 &= \underline{F}_4 \end{aligned}$$

$$\begin{aligned}
1 &\begin{cases} P_{11}^1 + P_{11}^2 + 0 + 0 + 0 = R_{11} \\ P_{12}^1 + P_{12}^2 + 0 + 0 + 0 = R_{12} \end{cases} \\
2 &\begin{cases} P_{21}^1 + 0 + P_{11}^3 + P_{11}^4 + 0 = R_{21} \\ P_{22}^1 + 0 + P_{12}^3 + P_{12}^4 + 0 = 0 \end{cases} \\
3 &\begin{cases} 0 + P_{21}^2 + P_{21}^3 + 0 + P_{11}^5 = 0 \\ 0 + P_{22}^2 + P_{22}^3 + 0 + P_{12}^5 = -0.36 \end{cases} \\
4 &\begin{cases} 0 + 0 + 0 + P_{21}^4 + P_{21}^5 = 0.72 \\ 0 + 0 + 0 + P_{22}^4 + P_{22}^5 = 0 \end{cases}
\end{aligned}$$

Globale Gleichgewichtsbeziehung

$$
\begin{aligned}
P^1_{11} + P^2_{11} + 0 + 0 + 0 &= R_{11}\\
P^1_{12} + P^2_{12} + 0 + 0 + 0 &= R_{12}\\
P^1_{21} + 0 + P^3_{11} + P^4_{11} + 0 &= R_{21}\\
P^1_{22} + 0 + P^3_{12} + P^4_{12} + 0 &= 0\\
0 + P^2_{21} + P^3_{21} + 0 + P^5_{11} &= 0\\
0 + P^2_{22} + P^3_{22} + 0 + P^5_{12} &= 0.36\\
0 + 0 + 0 + P^4_{21} + P^5_{21} &= 0.72\\
\underbrace{0}_{\hat{P}^1} + \underbrace{0}_{\hat{P}^2} + \underbrace{0}_{\hat{P}^3} + \underbrace{P^4_{22}}_{\hat{P}^4} + \underbrace{P^5_{22}}_{\hat{P}^5} &= \underbrace{0}_{\hat{F}}
\end{aligned}
$$

$$\hat{\underline{u}} = \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$

$$\hat{\underline{P}}^1 + \hat{\underline{P}}^2 + \hat{\underline{P}}^3 + \hat{\underline{P}}^4 + \hat{\underline{P}}^5 = \hat{\underline{F}} \quad \Longleftrightarrow \text{"System im Gleichgewicht"}$$

$$\hat{\underline{K}}^1 \hat{\underline{u}} + \hat{\underline{K}}^2 \hat{\underline{u}} + \hat{\underline{K}}^3 \hat{\underline{u}} + \hat{\underline{K}}^4 \hat{\underline{u}} + \hat{\underline{K}}^5 \hat{\underline{u}} = \hat{\underline{F}}$$

$$\left(\hat{\underline{K}}^1 + \hat{\underline{K}}^2 + \hat{\underline{K}}^3 + \hat{\underline{K}}^4 + \hat{\underline{K}}^5\right) \hat{\underline{u}} = \hat{\underline{F}}$$

Globale Steifigkeits-matrix → $\hat{\underline{K}}\ \hat{\underline{u}} = \hat{\underline{F}}$

Globale Elementmatrizen

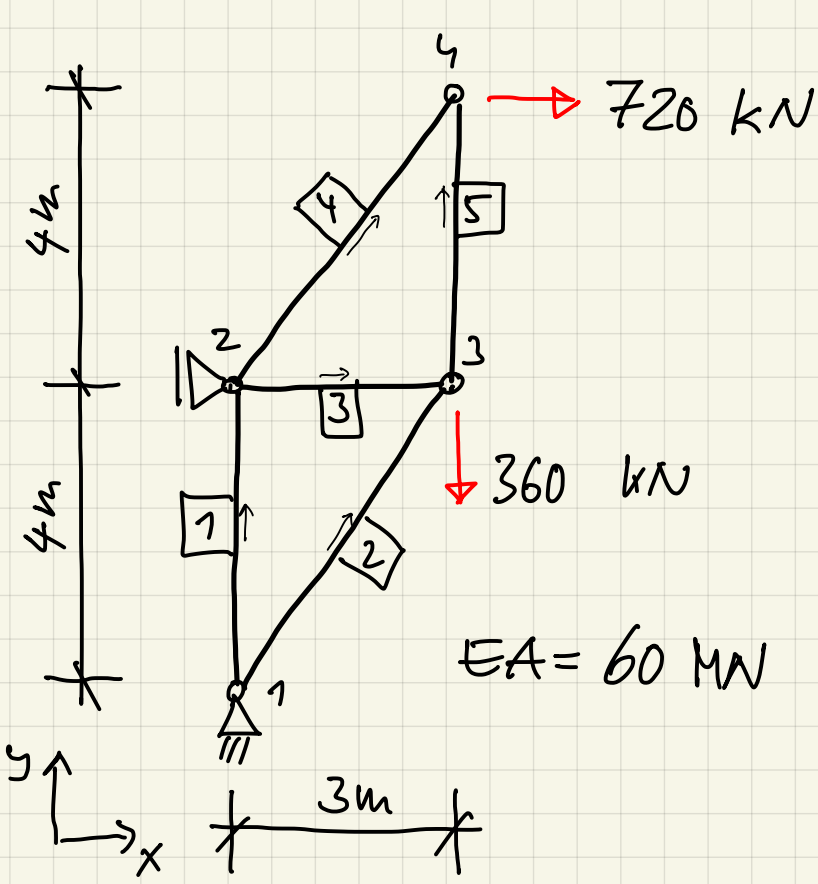
$$\hat{\underline{K}}^1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

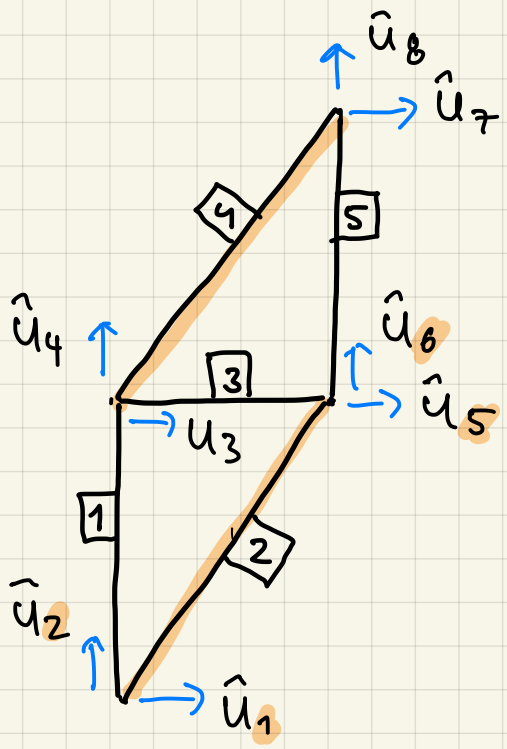
$$\hat{\underline{K}}^5 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -15 & 0 & 15 \end{bmatrix}$$

$$\hat{\underline{K}}^2 = \begin{bmatrix} 4.32 & 5.76 & 0 & 0 & -4.32 & -5.76 & 0 & 0 \\ 5.76 & 7.68 & 0 & 0 & -5.76 & -7.68 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -4.32 & -5.76 & 0 & 0 & 4.32 & 5.76 & 0 & 0 \\ -5.76 & -7.68 & 0 & 0 & 5.76 & 7.68 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\underline{K}}^3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 20 & 0 & -20 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -20 & 0 & 20 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\underline{K}}^4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4.32 & 5.76 & 0 & 0 & -4.32 & -5.76 \\ 0 & 0 & 576 & 7.68 & 0 & 0 & -5.76 & -7.68 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -4.32 & -5.76 & 0 & 0 & 4.32 & 5.76 \\ 0 & 0 & -5.76 & -7.68 & 0 & 0 & 576 & 7.68 \end{bmatrix}$$



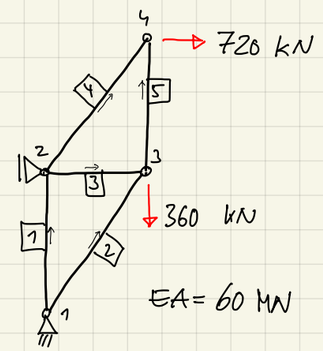


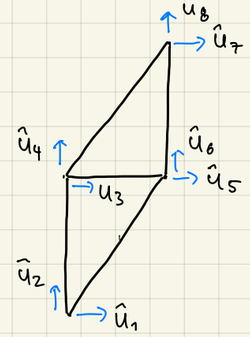
$EA = 60\ MN$

$$\begin{bmatrix} \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix} = \begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ \begin{bmatrix} & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix} \end{array} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$

Lineares Gleichungssystem

$$
\begin{bmatrix}
4.32 & 5.76 & 0 & 0 & -4.32 & -5.76 & 0 & 0 \\
5.76 & 22.68 & 0 & -15 & -5.76 & -7.68 & 0 & 0 \\
0 & 0 & 24.32 & 5.76 & -20 & 0 & -4.32 & -5.76 \\
0 & -15 & 5.76 & 22.68 & 0 & 0 & -5.76 & -7.68 \\
-4.32 & -5.76 & -20 & 0 & 24.32 & 5.76 & 0 & 0 \\
-5.76 & -7.68 & 0 & 0 & 5.76 & 22.68 & 0 & -15 \\
0 & 0 & -4.32 & -5.76 & 0 & 0 & 4.32 & 5.76 \\
0 & 0 & -5.76 & -7.68 & 0 & -15 & 5.76 & 22.68
\end{bmatrix}
\begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}
= \begin{bmatrix} R_{11} \\ R_{12} \\ R_{21} \\ 0 \\ 0 \\ -0.36 \\ 0.72 \\ 0 \end{bmatrix}
= 0\,!
$$





$$
\begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & -15 & 5.76 & 22.68 & 0 & 0 & -5.76 & -7.68 \\
-4.32 & -5.76 & -20 & 0 & 24.32 & 5.76 & 0 & 0 \\
-5.76 & -7.68 & 0 & 0 & 5.76 & 22.68 & 0 & -15 \\
0 & 0 & -4.32 & -5.76 & 0 & 0 & 4.32 & 5.76 \\
0 & 0 & -5.76 & -7.68 & 0 & -15 & 5.76 & 22.68
\end{bmatrix}
\begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}
= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -0.36 \\ 0.72 \\ 0 \end{bmatrix}
$$

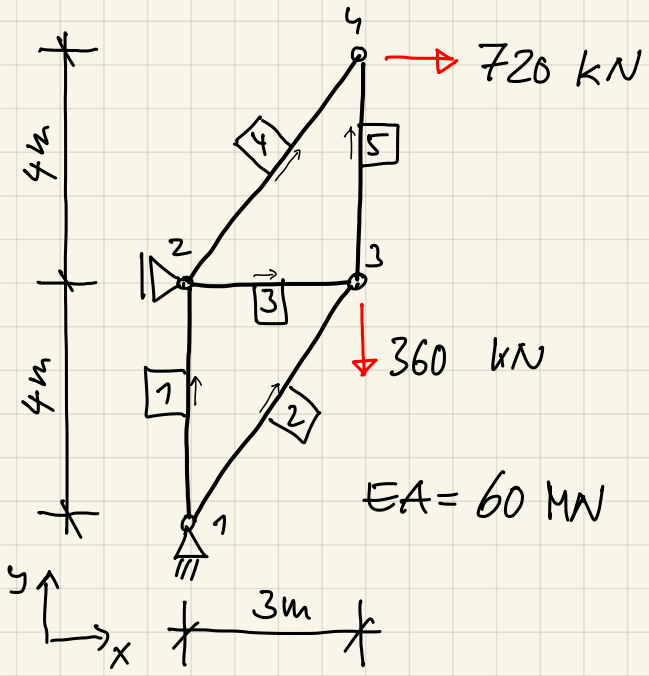
Lösung LGS

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -15 & 5.76 & 22.68 & 0 & 0 & -5.76 & -7.68 \\ -4.32 & -5.76 & -20 & 0 & 24.32 & 5.76 & 0 & 0 \\ -5.76 & -7.68 & 0 & 0 & 5.76 & 22.68 & 0 & -15 \\ 0 & 0 & -4.32 & -5.76 & 0 & 0 & 4.32 & 5.76 \\ 0 & 0 & -5.76 & -7.68 & 0 & -15 & 5.76 & 22.68 \end{bmatrix} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -0.36 \\ 0.72 \\ 0 \end{bmatrix}$$

$$\underline{\hat{u}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.064 \\ 0.0495 \\ -0.209 \\ 0.616 \\ 0.273 \end{bmatrix}$$

$$\underline{u}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \underline{u}_2 = \begin{bmatrix} 0 \\ 0.64 \end{bmatrix}, \quad \underline{u}_3 = \begin{bmatrix} 0.0495 \\ -0.209 \end{bmatrix}, \quad \underline{u}_4 = \begin{bmatrix} 0.616 \\ 0.273 \end{bmatrix}$$

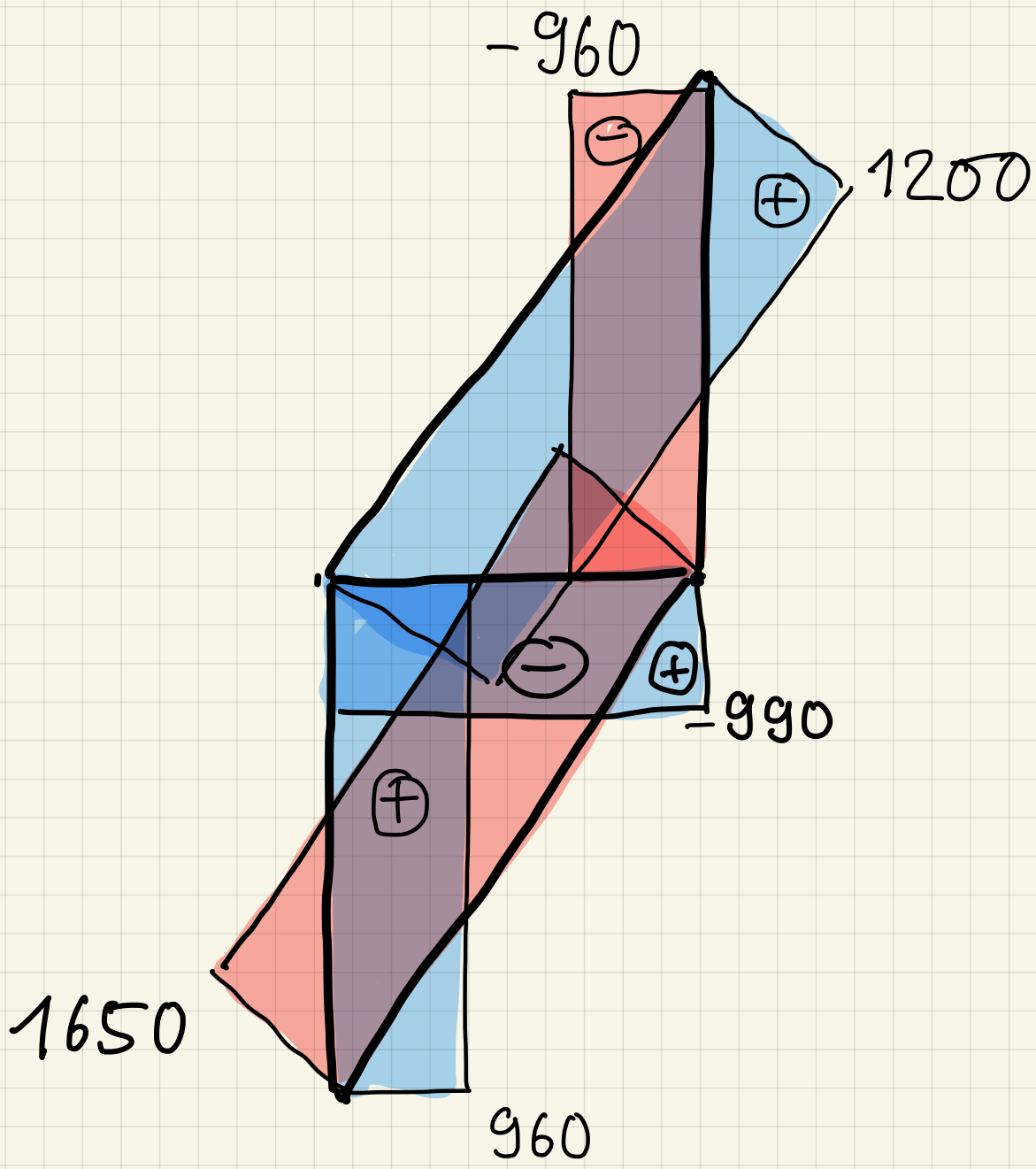
Schnittgrößen



Allgemein

$$N = \frac{EA}{l_0} \left(\underline{u}_2 \cdot \underline{d}_0 - \underline{u}_1 \cdot \underline{d}_0 \right)$$

$$N = \frac{EA}{l_0} \cdot \left(\underline{u}_2 - \underline{u}_1 \right) \cdot \underline{d}_0$$



N [kN]

Vektor der globalen Stabendkräfte für Stab 1

$$\begin{bmatrix} P^1_{11} \\ P^1_{12} \\ P^1_{21} \\ P^1_{22} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$

$$\Leftrightarrow \quad \underline{\hat{P}}^1 = \underline{\hat{K}}^1 \, \underline{\hat{u}}$$

Globale Steifigkeitsmatrix für Stab 1 $\underline{\hat{K}}^1 = (\underline{A}^1)^T \, \underline{K}^1 \, \underline{A}^1$

$$\begin{bmatrix} P^1_{11} \\ P^1_{12} \\ P^1_{21} \\ P^1_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 \end{bmatrix} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \end{bmatrix}$$

$L^1_0 = 4$, $EA = 60$, $\underline{d}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\underline{t}_0 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$

$$\underline{K}^1 = \frac{60}{4} \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 \end{bmatrix}$$

$$\begin{bmatrix} P^1_{11} \\ P^1_{12} \\ P^1_{21} \\ P^1_{22} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$

$$\underline{\hat{P}}^1 = \underbrace{\left(\underline{A}^1\right)^T \underline{K}^1 \underline{A}^1}_{\hat{\underline{K}}^1} \underline{\hat{u}} = \hat{\underline{K}}^1 \underline{\hat{u}}$$

$$\begin{bmatrix} P^1_{11} \\ P^1_{12} \\ P^1_{21} \\ P^1_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 \end{bmatrix} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \end{bmatrix}$$

$$\begin{bmatrix} P^1_{11} \\ P^1_{12} \\ P^1_{21} \\ P^1_{22} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P^1_{11} \\ P^1_{12} \\ P^1_{21} \\ P^1_{22} \end{bmatrix}$$

$$\begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$

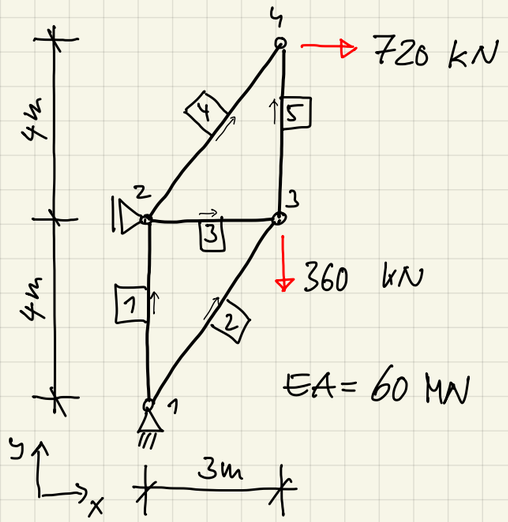
$$\hat{\underline{K}}^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

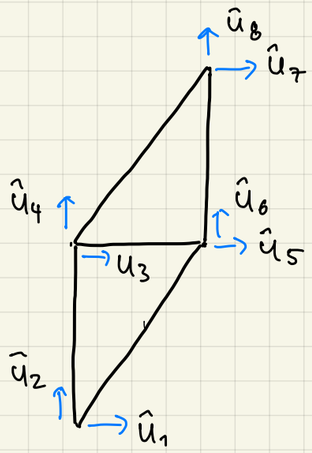
$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 \\ 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 15 & 0 & -15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -15 & 0 & 15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Assemblierungsmatrix A

Für Stab 2

$$\underline{I}^e = \begin{bmatrix} 1 \\ 2 \\ 5 \\ 6 \end{bmatrix} \rightsquigarrow \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_5 \\ \hat{u}_6 \end{bmatrix} = \begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array} \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \hat{u}_3 \\ \hat{u}_4 \\ \hat{u}_5 \\ \hat{u}_6 \\ \hat{u}_7 \\ \hat{u}_8 \end{bmatrix}$$





Allgemein

$$A^e_{ij} = \begin{cases} 1 & \text{falls } I^e_i = j \\ 0 & \text{sonst} \end{cases}, \quad i = 1, \dots, 4, \quad j = 1, \dots, 2 \cdot N_n$$

Globale Steifigkeitsmatrix zu Stab e

$$\hat{\underline{K}}^e = (\underline{A}^e)^T \underline{K}^e \underline{A}^e \qquad \hat{K}^e_{I^e_i I^e_j} = K^e_{ij}$$

→ Umsortierung!

Berechnung der Verschiebungen

Wir fassen die Schritte aus den vorangegangenen Abschnitten nochmal allgemein zusammen:

1. Nützliche Variablen:

   N_e (Anzahl Stäbe), N_n (Anzahl Knoten), $N = 2N_n$ (Anzahl Freiheitsgrade)

2. Globale Steifigkeitsmatrix $\hat{\mathbf{K}}$ der Größe $N \times N$ mit Nullen vorbelegen

3. Schleife über alle Stäbe
   - Variablen: n_1, n_2 (Knotennummern), $\mathbf{x}_1, \mathbf{x}_2$ (Knotenkoordinaten)
   - Elementsteifigkeitsmatrix $\mathbf{K}$ berechnen
   - Indexvektor $I = (2n_1 - 1, 2n_1, 2n_2 - 1, 2n_2)$ für Element aufstellen
   - Matrix $\mathbf{K}$ mit Indexvektor I auf globale Matrix $\hat{\mathbf{K}}$ aufaddieren
4. Äußere Lasten in Vektor $\hat{\mathbf{F}}$ anordnen
5. Auflagerbedingungen einbauen
6. Lineares Gleichungssystem lösen